

LDE with R side non zero.

Solution: $y(x) = y_c + y_p \rightarrow$ Particular Integral
↳ Complementary function

✓ ① $\frac{d^2 y}{dx^2} + 9y = 5x^2$ — (1)

Let $y = e^{mx}$ be a possible solution of

$$\frac{d^2 y}{dx^2} + 9y = 0$$

$$A.E: m^2 + 9 = 0$$

$$\Rightarrow m = \pm i3$$

$$y_c = A \cos 3x + B \sin 3x$$

$$\text{Let } D = \frac{d}{dx},$$

$$D^2 = \frac{d^2}{dx^2}$$

$$\textcircled{1} \Rightarrow D^2 y + 9y = 5x^2$$

$$\Rightarrow y(D^2 + 9) = 5x^2$$

$$y_p = \frac{1}{D^2 + 9} 5x^2$$

$$= \frac{1}{9} \frac{1}{\left(1 + \frac{D^2}{9}\right)} 5x^2$$

$$= \frac{1}{9} \left(1 + \frac{D^2}{9}\right)^{-1} 5x^2$$

$$= \frac{1}{9} \left(1 - \frac{D^2}{9} + \frac{D^4}{81} - \dots\right) 5x^2$$

$$= \frac{1}{9} \left(5x^2 - \frac{10}{3}\right)$$

$$= \frac{5x^2}{9} - \frac{10}{81}$$

$$\therefore y(x) = y_c + y_p$$

$$= A \cos 3x + B \sin 3x + \frac{5x^2}{9} - \frac{10}{81}$$

(2)

$$\frac{dy}{dx} - 2 \frac{dy}{dx} + 3x = 3x + 2, \quad y(0) = \frac{1}{2}, \quad y'(0) = \frac{4}{9}$$

$$\textcircled{2} \quad y_p = \frac{1}{f(a)} e^{ax} \quad \left| \begin{array}{l} \text{if } f(a) \neq 0 \\ \text{if } f(a) = 0 \end{array} \right.$$

$$= \frac{1}{f(a)} e^{ax}$$

$$\text{if } f(a) = 0,$$

$$y_p = x \frac{1}{f'(a)} e^{ax}$$

$$\text{if } f'(a) = 0$$

$$y_p = x^2 \frac{1}{f''(a)} e^{ax}$$

$$\sqrt{\frac{dy}{dx} - 3 \frac{dy}{dx} + 2y = e^{5x}} \quad \textcircled{1}$$

$$y(x) = C_1 e^x + C_2 e^{2x}$$

$$\text{if } \textcircled{1} \Rightarrow Dy - 3Dy + 2y = e^{5x}$$

$$\Rightarrow (D^2 - 3D + 2)y = e^{5x}$$

$$\Rightarrow y_p = \frac{1}{D^2 - 3D + 2} e^{5x}$$

$$= \frac{e^{5x}}{12}$$

$$\therefore y(x) = y_c + y_p = C_1 e^x + C_2 e^{2x} + \frac{1}{12} e^{5x}$$

$$(D^3 - 2D^2 - 5D + 6)y = e^{5x}$$

(3)

(2)

$$y_p = \frac{1}{f(D^2)} \sin ax$$

$$= \frac{1}{f(-a^2)} \sin ax$$

$$f(-a^2) = 0$$

$$\Rightarrow y_p = x \frac{1}{f'(D)} \sin ax$$

$$(D^2 - 3D + 4)y = \cos(4x+5) \quad \checkmark \quad Dy - 2Dy + y = \sin x \quad \sin 2x$$

$$\text{Conditions } y(0) = 1$$

$$y'(0) = 1$$

$$y_p = \frac{1}{D^2 - 3D + 4} \sin(4x+5)$$

$$= \frac{1}{-D^2 - 3D + 4} \sin(4x+5)$$

$$= \frac{1}{-12 - 3D} \sin(4x+5)$$

$$= \frac{1}{-(D+4)3} \sin(4x+5)$$

$$= - \frac{D-4}{D-4} \sin(4x+5)$$

$$= - \frac{D-4}{12} \sin(4x+5)$$

$$= - \frac{1}{12} \{ D \sin(4x+5) - 4 \sin(4x+5) \}$$

$$= - \frac{1}{12} \{ 4 \cos(4x+5) - 4 \sin(4x+5) \}$$

$$y_c = m^2 - 3m + 4$$

$$m = \frac{3 \pm \sqrt{9-16}}{2}$$

$$= \frac{3 \pm i\sqrt{7}}{2}$$

$$y_c = e^{3/2x} (A \cos \sqrt{7/2}x + B \sin \sqrt{7/2}x)$$

(4)

(1)

$$y_p = \frac{1}{f(D)} e^{ax} v$$

$$= e^{ax} \frac{1}{f(D+a)} v$$

(2)

$$y_p = \frac{1}{f(D)} x^n \cos ax, \quad \frac{1}{f(D)} x^n \sin ax$$

$$X = \frac{1}{f(D)} x^n \cos ax \quad Y = \frac{1}{f(D)} x^n \sin ax$$

$$X + iY = \frac{1}{f(D)} x^n (\cos ax + i \sin ax)$$

$$= \frac{1}{f(D)} x^n e^{iax}$$

(3)

$$y_p = \frac{1}{f(D)} xu$$

$$= x \frac{1}{f(D)} v - \frac{f'(D)}{[f(D)]^2} v$$

$$= e^{aix} \frac{1}{f(D+ai)} x^n$$

(1) $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = x^2 e^{2x}$

$$y(x) = (C_1 + C_2 x) e^{2x} + \frac{1}{12} x^4 e^{2x}$$

(11) $(D^2 - 1)y = x^2 \cos x$

$$y_c = C_1 e^x + C_2 e^{-x}$$

$$X = \frac{x^2 \cos x}{D^2 - 1} \quad Y = \frac{x^2 \sin x}{D^2 - 1}$$

$$X + iY = \frac{x^2 (\cos x + i \sin x)}{D^2 - 1}$$

$$= \frac{x^2 e^{ix}}{(D^2 - 1)}$$

$$= \frac{x^2 e^{ix}}{(D^2 - 1)}$$

$$= e^{ix} \frac{1}{(D^2 - 1)} x^2$$

$$= e^{ix} \frac{1}{D^2 + 2Di - 2} x^2$$

$$X = -\frac{1}{2} \{ (x^2 - 1) \cos x - 2x \sin x \}$$

$$Y = -\frac{1}{2} \{ 2x \cos x + (x^2 - 1) \sin x \}$$

$$\frac{d}{dx} \frac{1}{D^2 - 1} x^2 \cos x =$$

$$-\frac{1}{2} \{ (x^2 - 1) \cos x - 2x \sin x \}$$

$$y(x) = y_c + y_p$$

$$= \frac{e^{ix}}{-2} \frac{1}{(1 - Di - D^2/2)} x^2$$

$$= -\frac{1}{2} e^{ix} \{ 1 - (Di + D^2/2) \}^{-1} x^2$$

$$= -\frac{1}{2} e^{ix} \{ 1 + Di + D^2/2 + (iD + D^2/2)^2 + \dots \} x^2$$

$$= \frac{1}{2} e^{iax} \{ x^2 + 2xi + \underline{1} \}$$

$$= -\frac{1}{2} e^{iax} (x^2 + 2xi)$$

$$= -\frac{1}{2} (\cos x + i \sin x) (x^2 - 1 + 2xi)$$

$$\frac{dy}{dx} - 2\frac{dy}{dx} + y = x e^x \sin x \quad (1)$$

$$(D^2 - 2D + 1)y = x e^x \sin x$$

$$D = \frac{d}{dx}$$

$$y_c = (C_1 + C_2 x) e^x$$

$$y_p = (1) (D^2 - 2D + 1)y = x e^x \sin x$$

$$\Rightarrow y_p = \frac{1}{D^2 - 2D + 1} x e^x \sin x$$

$$= \frac{1}{(D-1)^2} x e^x (2 \sin x)$$

$$= e^x \frac{1}{(D+1)^2} 2 \sin x$$

$$= e^x \frac{1}{D^2} 2 \sin x$$

$$= e^x \left[2 \frac{1}{D^2} \sin x - \frac{2D}{(D^2)^2} \sin x \right]$$

$$= e^x [-2 \sin x - 2 \cos x]$$

$$= -e^x (2 \sin x + 2 \cos x)$$

$$y(x) = y_c + y_p$$

$$(D^2 - 2D + 1)y = x \sin x$$

$$\therefore y_p = \frac{1}{D^2 - 2D + 1} x \sin x$$

$$(1) (D^2 - 2D + 1)y = x e^x \sin x \quad (3)$$

let $y = e^{mx}$ be a solution
of $(D^2 - 2D + 1)y = 0$

$$\Rightarrow (D^2 - 2D + 1)e^{mx} = 0$$

$$\Rightarrow m^2 e^{mx} - 2m e^{mx} + e^{mx} = 0$$

$$\Rightarrow (m^2 - 2m + 1)e^{mx} = 0$$

$$\Rightarrow m^2 - 2m + 1 = 0$$

$$\Rightarrow (m-1)^2 = 0 \quad e^{mx} \neq 0$$

$$\Rightarrow (m-1)(m-1)$$

$$\Rightarrow m = 1, 1.$$

$$\therefore y_c = (C_1 + C_2 x) e^x$$

$$y_p = \frac{1}{D-\alpha} R(x) = e^{\alpha x} \int e^{-\alpha x} R(x) dx$$

$$R(x) = \sec x \tan x \csc x \cot x$$

$$y_p = \frac{1}{D+\alpha} R(x) = e^{-\alpha x} \int e^{\alpha x} R(x) dx$$

$$4(D^2 + a^2)y = \sec x$$

Let $y = e^{mx}$ a possible solution of $(D^2 + a^2)y = 0$ - (1)

$$m^2 + a^2 = 0, \quad y_c(x) = A \cos ax + B \sin ax$$

$$(1) \Rightarrow y_p = \frac{1}{D^2 + a^2} \sec x$$

$$= \frac{1}{(D+ia)(D-ia)} \sec x$$

$$= \frac{1}{2ai} \left[\frac{1}{D-ia} - \frac{1}{D+ia} \right] \sec x$$

$$= \frac{1}{2ai} \left[\frac{1}{D-ia} \sec x - \frac{1}{D+ia} \sec x \right]$$

$$= \frac{1}{2ai} \left[\int e^{iax} \frac{1}{e^{-iax}} \sec x dx - \int e^{-iax} \frac{1}{e^{iax}} \sec x dx \right]$$

$$= \frac{1}{2ai} \left[\int e^{iax} (\cos x - i \sin x) \frac{1}{\cos x} dx - \int e^{-iax} (\cos x + i \sin x) \frac{1}{\cos x} dx \right]$$

$$= \frac{1}{2ai} \left[\int e^{iax} (1 - i \tan x) dx - \int e^{-iax} (1 + i \tan x) dx \right]$$

$$= \frac{1}{2ai} \left[\int e^{iax} (1 - i \tan x) dx - \int e^{-iax} (1 + i \tan x) dx \right]$$

$$= \frac{1}{2ai} \left[x(e^{iax} - e^{-iax}) + \frac{1}{a} (e^{iax} + e^{-iax}) \ln |\cos x| \right]$$

$$= \frac{x}{2} \sin ax + \frac{1}{a^2} (\cos ax) \ln |\cos ax|$$

$$y(x) = y_c + y_p$$

(7) L.D.E with variable coefficient. (8) (V)

$$x = e^z \Rightarrow z = \ln x$$

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} = \frac{1}{x} \frac{dy}{dz}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dz} = Dy$$

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dz} - \frac{1}{x} \frac{dy}{dz}$$

$$x \frac{dy}{dx} = \frac{dy}{dz} - \frac{dy}{dz} = Dy - Dy = D(D-1)y$$

$$\text{Similarly, } x^2 \frac{d^2 y}{dx^2} = D(D-1)(D-2)y$$

$$(1) \quad x^3 \frac{d^3 y}{dx^3} + 3x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$$

$$D(D-1)(D-2)y + 3D(D-1)y - 2Dy + 2y = 0$$

$$\Rightarrow (D^3 - 3D + 2)y = 0 \quad (1)$$

$$\text{Let } y = e^{mz} \quad (2)$$

$$\text{A.E. } m^3 - 3m + 2 = 0$$

$$m = 1, 1, -2,$$

$$y(z) = e^z (c_1 + z c_2) + c_3 e^{-2z}$$

$$x^2 y'' - 3xy' + y = (c_1 + c_2 \ln x)x + c_3/x$$

$$(11) = x + z^2 \ln x$$

$$= e^z + z^2 e^{2z}$$

$$y_p = \frac{1}{(D-2)^2} e^z + z e^{2z}$$

$$m^3 y'' + 3m^2 y' + y = x + \ln x$$

$$(1+2x)^2 \frac{d^2 y}{dx^2} - 6(1+2x) \frac{dy}{dx} + 16y = 8(1+2x)^2 \quad \text{--- (1)}$$

Let $(1+2x) = e^z$ then $z = \ln(1+2x)$

$$\therefore \frac{dz}{dx} = \frac{2}{1+2x}$$

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} = \frac{dy}{dz} \cdot \frac{2}{1+2x}$$

$$\Rightarrow (1+2x) \frac{dy}{dx} = 2 \frac{dy}{dz} = 2Dy \quad \text{where } D = \frac{d}{dz}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{2}{1+2x} \cdot \frac{dy}{dz} \right) = \frac{2}{(1+2x)^2} \frac{d^2 y}{dz^2} - \frac{2}{(1+2x)^2} \frac{dy}{dz}$$

$$\Rightarrow (1+2x)^2 \frac{d^2 y}{dx^2} = 4 \frac{d^2 y}{dz^2} - 4 \frac{dy}{dz} = 4D(D-1)y$$

$$\text{(1)} \Rightarrow 4D(D-1)y - 6 \cdot 2Dy + 16y = 8(e^z)^2$$

$$\Rightarrow (D^2 - 4D + 4)y = 2e^{2z} \quad \text{--- (2)}$$

Let $y = e^{mz}$ be a possible solution of $(D^2 - 4D + 4)y = 0$

$$\text{A.E.: } m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0$$

$$m = 2, 2$$

$$y_c = (C_1 + C_2 z) e^{2z} = [C_1 + C_2 \ln(1+2x)] (1+2x)^2$$

$$\text{From (2), } y_p = \frac{1}{D^2 - 4D + 4} 2e^{2z}$$

$$= 2 \frac{1}{2D - 4} 2e^{2z}$$

$$= 2^2 \frac{1}{2} 2e^{2z}$$

$$= \frac{1}{2} 2^2 e^{2z}$$

$$= (1+2x)^2 \{ \ln(1+2x) \}^2$$

$$\therefore y(x) = y_c + y_p$$

$$= [C_1 + C_2 \ln(1+2x)] (1+2x)^2 + (1+2x)^2 \{ \ln(1+2x) \}^2$$

Bernoulli's Equation:

$$\frac{dy}{dx} + py = ay^n \text{ --- (1)}$$

p and q are function of x.

Derivation:

$$\frac{dy}{dx} + py = ay^n$$

$$\Rightarrow \frac{1}{y^n} \frac{dy}{dx} + \frac{py}{y^n} = a$$

$$\Rightarrow y^{-n} \frac{dy}{dx} + y^{-n+1} p = a \text{ --- (1)}$$

$$\text{Let } y^{-n+1} = v$$

$$(n+1) y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx}$$

$$\textcircled{1} \Rightarrow \frac{1}{1-n} \frac{dv}{dx} + pv = a \text{ --- (2)}$$

$$\Rightarrow \frac{dv}{dx} + (1-n)p v = a(1-n) \text{ --- (2)}$$

$$\text{I.F} = e^{\int (1-n)p dx}$$

$$\textcircled{2} \times \text{I.F} \Rightarrow e^{\int (1-n)p dx} \frac{dv}{dx} + (1-n)p v e^{\int (1-n)p dx} = (1-n)a e^{\int (1-n)p dx}$$

$$\Rightarrow \frac{d}{dx} [v e^{\int (1-n)p dx}] = (1-n)a e^{\int (1-n)p dx}$$

$$\Rightarrow v e^{\int (1-n)p dx} = \int (1-n)a e^{\int (1-n)p dx} dx + C$$

This is the reversed solution of Bernoulli's eqn.

$$\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos 2y \quad \text{--- (1)}$$

$$\frac{1}{\cos 2y} \frac{dy}{dx} + \frac{1}{x} \frac{2 \sin y \cos y}{\cos 2y} = x^3$$

$$\Rightarrow \sec 2y \frac{dy}{dx} + \frac{2}{x} \tan y = x^3 \quad \text{--- (2)}$$

let $\tan y = v$

$$\sec 2y \frac{dy}{dx} = \frac{dv}{dx}$$

$$\sec 2y \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{(2)} \Rightarrow \frac{dv}{dx} + \frac{2}{x} v = x^3 \quad \text{--- (3)}$$

I.F. of eqn (3), I.F. = $e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2$

Multiplying (3) by I.F. we get,

$$\frac{d}{dx} (v x^2) = x^5$$

$$\Rightarrow v x^2 = \int x^5 dx + c$$

$$\Rightarrow \tan 2y = \frac{x^6}{6} + c$$

$$\frac{dy}{dx} + \frac{xy}{1-x^2} = x\sqrt{y} \quad \text{--- (1)}$$

$$y^{-1/2} \frac{dy}{dx} + \frac{x}{1-x^2} y^{1/2} = x \quad \text{--- (2)}$$

let $y^{1/2} = v$

$$\frac{1}{2} y^{-1/2} \frac{dy}{dx} = \frac{dv}{dx}$$

$$y^{1/2} \frac{dy}{dx} = 2 \frac{dv}{dx}$$

$$\text{(2)} \Rightarrow \frac{2 dv}{dx} + \frac{x}{1-x^2} v = x$$

$$\frac{dv}{dx} + \frac{x}{2(1-x^2)} v = \frac{x}{2}$$

$$\begin{aligned} \text{I.F.} &= e^{\int \frac{x}{2(1-x^2)} dx} = e^{-\frac{1}{4} \int \frac{-2x}{1-x^2} dx} \\ &= e^{-\frac{1}{4} \ln(1-x^2)} \\ &= e^{\ln(1-x^2)^{-1/4}} \\ &= \frac{1}{(1-x^2)^{1/4}} \end{aligned}$$

$$\therefore \frac{d}{dx} \frac{v}{(1-x^2)^{1/4}} = \frac{x}{2(1-x^2)^{1/4}}$$

$$\begin{aligned} \frac{v}{(1-x^2)^{1/4}} &= \int \frac{x}{2(1-x^2)^{1/4}} dx \\ &= -\frac{1}{4} \int \frac{-2x}{(1-x^2)^{1/4}} dx \end{aligned}$$

$$\frac{\sqrt{y}}{(1-x^2)^{1/4}}$$

$$= -\frac{1}{4} (1-x^2)^{3/4} + c$$